

Matthew Saunders

matt.saunders.963@gmail.com | 410-428-3544 | Baltimore, Maryland | [linkedin.com/in/mattsaundercs/](https://www.linkedin.com/in/mattsaundercs/)

Summary

Senior Computer Science student with extensive programming skills, hardware knowledge, scientific research skills, and a deep passion for space and aeronautical sciences. Thrives on solving complex problems using analytical and critical thinking skills while always learning and staying curious. Team oriented with excellent communication and collaboration skills. Striving for a lifelong career involving theoretical and practical space sciences and exploration to benefit humanity for the future. Ready to continue my academic career and pair my Computer Science knowledge with Aerospace Engineering.

Education

University of Maryland Global Campus, Bachelor of Science in Computer Science, Spring 2027

Relevant Coursework: Discrete Structures, Algorithm Design, Advanced Programming, OOP, Compiler Theory, ML, Calculus I, II, III, Linear Algebra, Differential Equations, Algebraic Structures, Real Analysis, Complex Analysis

University of Colorado, Boulder, Bachelor of Science in Computer Science, August 2022 – April 2023

Relevant Coursework: Astronomy, Ethics, Philosophy, General Education Credits

Sollers Point Technical High School, Technical Certificate in Computer Science, September 2017 – May 2021

Relevant Coursework: Computer Architecture, C, Java, Object Oriented Programming, Microcontrollers, Networking

Sparrows Point High School, High School Diploma, September 2017 – May 2021

Skills

Languages: C, C++, C#, ASM, Python, Swift, Java, JavaScript, TypeScript, SQL, HTML, CSS

Frameworks: React.js, Node.js, Next.js, SwiftUI, NumPy, Tesseract, Matplotlib, TensorFlow, PyTorch, Pandas

Developer Tools: Amazon Web Services, PostgreSQL, Oracle SQL, Docker, Firebase, GitHub, Jira, Flex, Make

Operating Systems: Windows, MacOS, Linux, Unix

Technical: Computer Hardware, Computer Software, MATLAB, Soldering, Embedded Programming, Networking

Projects

Mobile Computerized Automated Observatory: C, Python, Embedded Programming, Tesseract, Linux

- Newtonian reflector telescope built with an aluminum extrusion frame and a 4.5-inch spherical primary mirror.
- Powered by Pi Finder, uses a Raspberry Pi and motors in the base to track celestial objects across the sky.
- Main camera uses a Sony ZVE-10, which allows for a 6000 X 4000 resolution and multiple hour-long exposures.
- Easily movable by car. Maximum achieved magnification of 269x. Field of view 1.6 degrees.

NOAA Satellite APT Signal Decoder: C, Python, Embedded Programming, Linux, Networking, Soldering

- Custom-built device and Quadrifilar Helix Antenna that listens for APT signals from passing NOAA weather satellites.
- Built using a Raspberry Pi for the computer and an SDR for the radio signal capture. 3D printed enclosure.
- When a signal is found, the device begins automatic capture and processes it at the proper sample rate.
- Processed signals get sent to WXtoIMG for decoding into full pictures. Software can label/outline landmasses as well as combine different sensors together to get visible or infrared images of earth.

Deployable Radio Beacon: Python, Embedded Programming, Linux, Soldering

- Portable, solar powered, and weatherproof radio beacon that could easily be deployed anywhere within seconds.
- Built using a microcontroller, an audio interface, a rechargeable battery, and a 10 watt ham radio for TX and RX.
- Interfaced using DTMF tones transmitted from another handheld radio. Different tones had different modes.
- Could transmit SSTV images taken from onboard camera. Could also transmit Morse code messages at a set interval.
- Other modes included voice repeater, beacon, and record mode, allowing it to be a versatile mobile radio hub.

GoDash iOS app: Swift, SwiftUI, MacOS, iOS

- iOS app that uses the phone's back camera and a hands free car mount to act as a dash cam while driving.
- App records up to 10 minutes of looping video that can be saved directly to the device with one click.
- Interface and UI employ distraction free multitasking using half screen modules for apps like maps and music.
- Apps within modules can be opened simultaneously to prevent app switching while driving.
- App applies an overlay onto dash cam clips that contains active telemetry such as G-forces, speed, and heading.

Work Experience

Micro Center, System Sales Associate, Parkville, Maryland, November 2024 – April 2025

- Managed full solution tech sales including hardware, software, and warranties to individuals and business clients.
- Created a needs analysis for every client to fit in with the system requirements and budget constraints.
- Worked in parallel with the IT team to allow seamless data transfers, system setups, and custom built systems.
- Maintained constant up to date knowledge on the newest advancements in technology. Ex. NPUs, ARM, AI.

BMW Group, Client Advisor and Lease Manager, Towson, Maryland, January 2023 – May 2024

- Controlled client acquisition, client relations, deal building, and closing of performance and high-line cars.
- Surpassed monthly goal of 12 deals and always maintained a client book of over 100 clients.
- Promoted to lease manager and increased recurring customer lease numbers by over 60% using equity mining tools.
- Created overseas orders and facilitated custom imports from Germany to the port of Baltimore.

Best Buy, Sales Associate, White Marsh, Maryland – Boulder, Colorado, November 2020 – May 2022

- Managed consumer sales of phones, computers, printers, TVs, and appliances.
- Excelled in selling Best Buy Total Tech on transactions by building rapport with customers.
- Communicated with outside companies for telecommunication service contracts and warranty repair orders.
- Obtained technology training, allowing me to work in every department Best Buy offers.

Certifications, Extracurriculars, and Volunteer Work

NASA's Open Science Essentials, obtained August 18, 2026

NASA's Open Science 101, obtained August 21, 2026

Boy Scouts of America, Star Scout, obtained 2017

Volunteer work: Scouting for Food, Wreaths Across America, State Park/roadway cleanup projects, Astronomy Teacher

The Planetary Society, Member, Mariner

Future Academic Plans

Master of Science in Aerospace Engineering, Planned start date, TBD

PhD in Computational Physics, Planned start date, TBD

Interests

Astrophotography, Aerospace, Computer Building, Camping, Hiking, Traveling, Rock Climbing, Music, Movies